Bayslope

INFRINGEMENT REPORT

SUBJECT PATENT: **US9455987B2**

PRIORITY DATE: **APRIL 17, 2014**

INFRINGING ENTITY NAME: **NETFLIX**

CHART STRENGTH: **HIGH**

PATENT TECHNOLOGY: **MEDIA STREAMING AND GEO-VERIFICATION**

Infringing Company: Netflix

Product name: Netflix Household

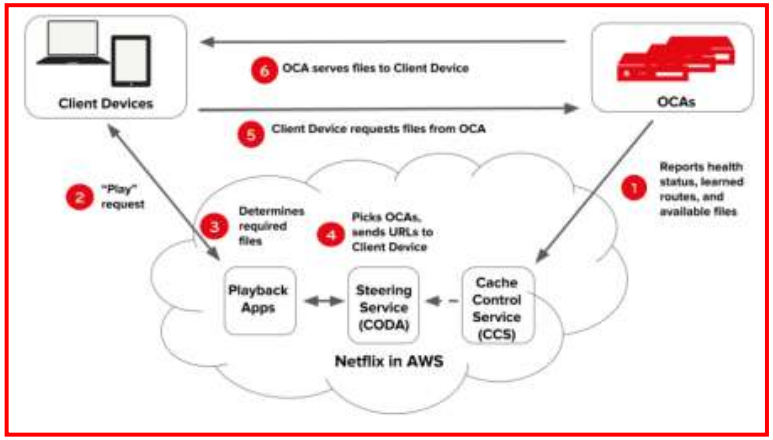
Netflix Revenue: $39 Billion

Company Type: Large Entity

Product Release date: 23 May 2023

US9455987B2	Netflix Household
1. A method of geo-verification for access to data stored in a memory of a server having a processor and a network interface connected to the memory, the method comprising:	***Analyst Comment:*** *A method of geo-verification (e.g., the IP address of Main Place) for access to data (e.g., movies or TV shows) stored in the memory of a server (e.g., AWS server) having a processor and a network interface connected to the memory.* *Netflix Household is a system that verifies whether a user is accessing Netflix from an authorized location. It checks IP addresses to verify if the device is within the designated household or verified location.* We use information such as IP addresses, device IDs, and account activity to determine whether a device signed into your account is part of your Netflix Household. Source: https://help.netflix.com/en/node/128339/us A Netflix account is meant to be shared by people who live together in one household. People who are not in your household will need to sign up for their own account to watch Netflix. You can manage who uses your account by setting a Netflix Household. A Netflix Household is a collection of the devices connected to the internet at the main place you watch Netflix. A Netflix Household can be set using a TV device. All other devices that use your Netflix account on the same internet connection as this TV will automatically be part of your Netflix Household. *Selection of the main place through a TV* Source: https://help.netflix.com/en/node/124925/us **Server**

	NETFLIX **Netflix on AWS** Netflix is one of the world's leading entertainment services with over 260 million members in more than 190 countries. Netflix uses AWS for nearly all its computing and storage needs, including databases, analytics, recommendation engines, video transcoding, and more—hundreds of functions that in total use more than 100,000 server instances on AWS. Netflix entertains the world, providing a wide variety of TV series, films and games to hundreds of millions of members across the globe in over 30 languages. Netflix builds diversity, inclusion, equity, and a global outlook into everything it does, and by fostering a culture of courage, empathy, and curiosity, Netflix can move faster to develop new stories and better ways of sharing them with its members around the world. Netflix relies on AWS to help it innovate with speed and consistently deliver best-in-class entertainment. AWS provides Netflix with compute, storage, and infrastructure that allow the company to scale quickly, operate securely, and meet capacity needs anywhere in the world. Moreover, Netflix, a leading content producer, has used AWS to build a studio in the cloud. This virtual studio enables Netflix to engage top artistic talent, no matter the location, and Netflix artists and partners have the freedom to collaborate without technological or geographical barriers. Source: https://aws.amazon.com/solutions/case-studies/innovators/netflix/ *Memory storage and processor*

	 *Network infrastructure* Source: https://openconnect.netflix.com/Open-Connect-Overview.pdf  *Data* Source: https://www.netflix.com/
receiving a selection of an approved location from the client computing device;	***Analyst Comment:*** *The method comprising receiving a selection of an approved location (e.g., main place) from the client computing device (e.g., TV, Mobile, etc.).*

determining whether a licensed number of approved geographic areas associated with the client computing device has been exceeded;	*Further, Netflix allows the user to update the approved location in case of determining whether a licensed number of approved geographic areas associated with the client computing device has been exceeded.* *Netflix Household enforces location-based access control. The user sets a primary location (approved geographic area) using a TV or another client device. Netflix continuously monitors access patterns by tracking IP addresses and device locations. If Netflix detects access from multiple geographic areas beyond the permitted limit, it prompts errors or verification requests, effectively determining when the number of approved geographic areas has been exceeded.* **Approved location** A Netflix account is meant to be shared by people who live together in one household. People who are not in your household will need to sign up for their own account to watch Netflix. You can manage who uses your account by setting a Netflix Household. A Netflix Household is a collection of the devices connected to the internet at the main place you watch Netflix. A Netflix Household can be set using a TV device. All other devices that use your Netflix account on the same internet connection as this TV will automatically be part of your Netflix Household. Source: https://help.netflix.com/en/node/124925/us *Selection of the main place through a TV* **How to set or update a Netflix Household** A Netflix Household is a collection of devices connected to the internet at the main place you watch Netflix. A Netflix Household can be set using most TV devices. Devices that use your Netflix account on the same internet connection as this TV will automatically be part of your Netflix Household.

Set or update your Netflix Household

You can set or update your Netflix Household when signed into Netflix from most TVs connected to your internet.

1. From the Netflix home screen on your TV, press left on your remote to open the menu.
2. Select **Get Help** > **Manage Netflix Household**.

   > **NOTE:** If you don't see **Manage Netflix Household** on the menu, you will need to use a different TV device.

3. Select **Update Netflix Household**.
4. Select **Send Email** or **Send Text**. A verification link will be sent to the account email address or phone number. Verification links expire after 15 minutes.

   > **NOTE:** If you have not added a phone number or email address to your account, you will only see one option.

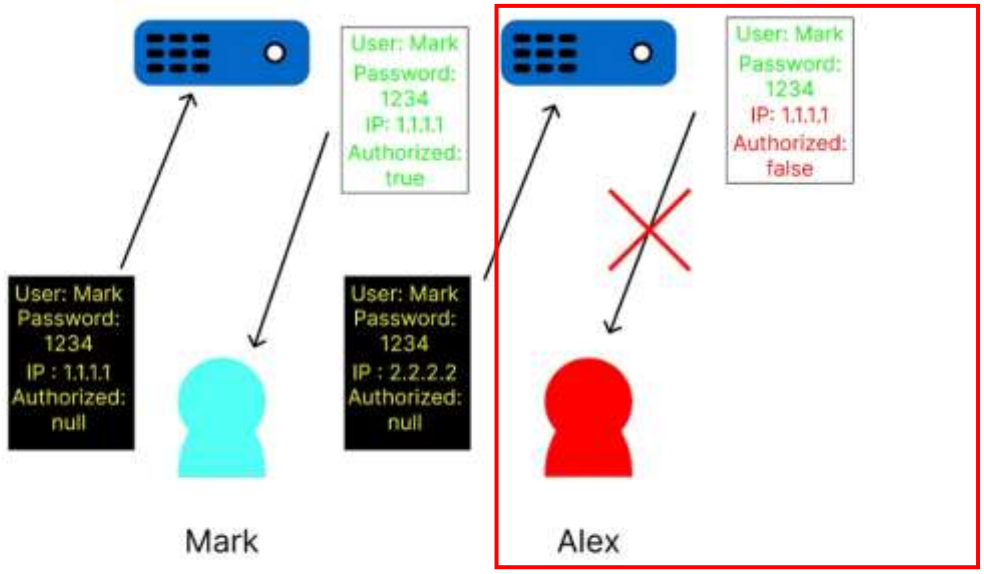
5. If you do not receive the verification link by email or text, click **Resend Email** or **Resend Text**. Or you can choose another option.
6. Select **Yes, This Was Me** in the email, or tap the link in the text message, then **Confirm Update** to continue.
7. You will receive a confirmation email and your TV will refresh so you can start watching.

We use information such as IP addresses, device IDs, and account activity to determine whether a device signed into your account is part of your Netflix Household.

Source: https://help.netflix.com/en/node/128339/us

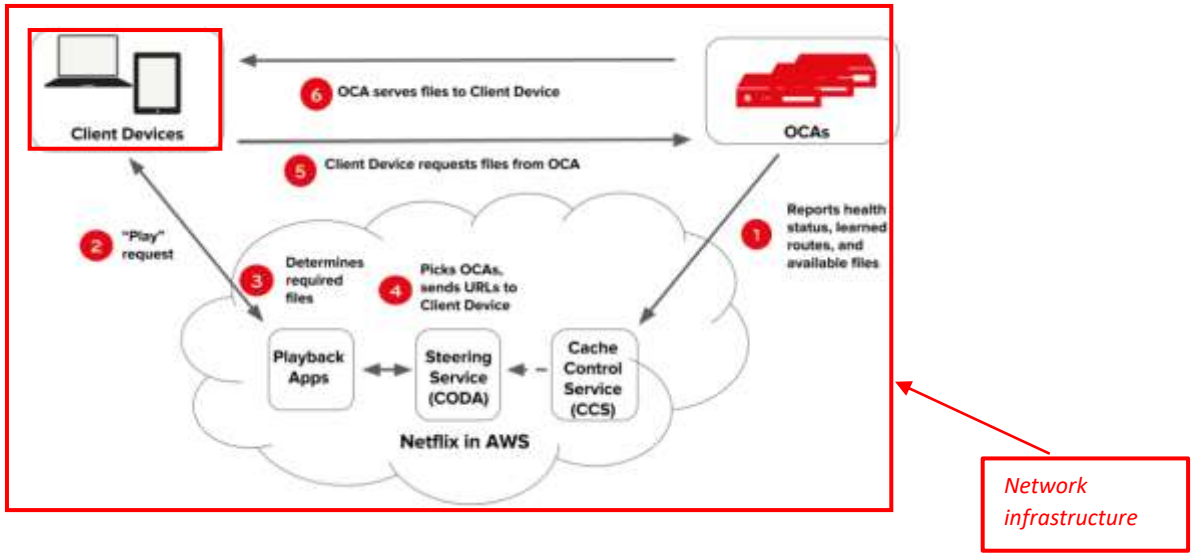
If you're using a mobile device or computer that's part of the Netflix Household but see this error, confirm that the device is connected to the same Wi-Fi as a TV or TV-connected device associated with the Netflix Household. If the problem persists, update the Netflix Household from your TV or TV-connected device and then restart the Netflix app on your mobile device or refresh the Netflix website on your computer.

Source: https://help.netflix.com/en/node/133821/us

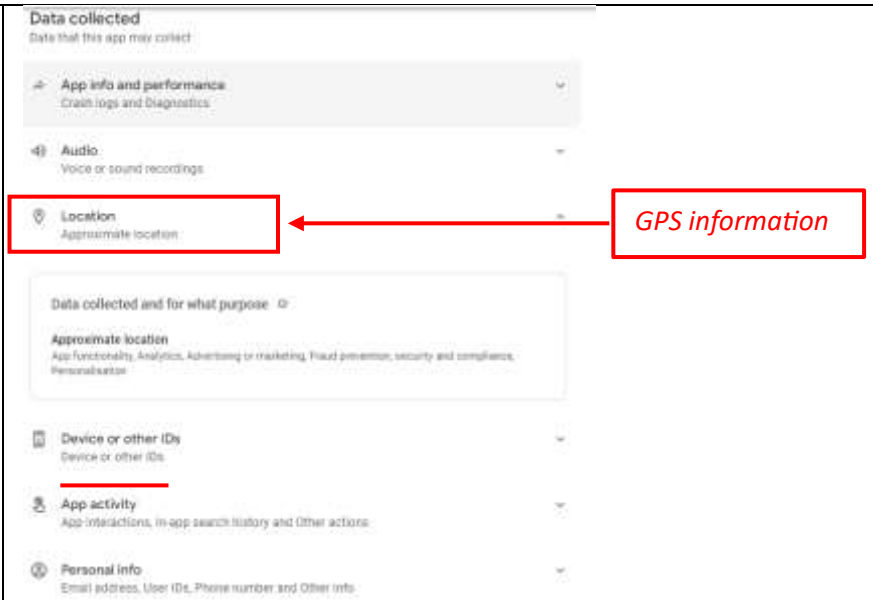
	 Figure 3. Netflix rejecting request with correct credentials because the IP address is not correct for the account. As we can see, it's not enough to be authenticated now. With the new policy, you need to make requests from a whitelisted IP address and may even need to have your device's MAC or private IP address whitelisted by your own account (device management is one of the "safety" and control features Netflix stressed in their pressers). Source: https://dpericich.medium.com/how-to-get-around-netflixs-new-household-policy-d2c41fa8ffd5
generating an approved geographic area based on the selected location; storing, in the memory, access control data including the approved geographic area;	***Analyst comment:*** *Generating an approved geographic area (e.g., General geographic location) based on the selected location; (e.g., main place).* *Storing, in the memory, access control data including the approved geographic area.* *When a user sets up Netflix Household, they select a primary location using a TV or another client device connected to their home network. Based on this selection, Netflix defines an approved*

	geographic area, typically by associating it with the home's IP address, Wi-Fi network, and device activity. This geographic area serves as the reference point for verifying future access attempts. **Set or update your Netflix Household** You can set or update your Netflix Household when signed into Netflix from most TVs connected to your internet. 1. From the Netflix home screen on your TV, press left on your remote to open the menu. 2. Select **Get Help** > **Manage Netflix Household**. > **NOTE:** If you don't see **Manage Netflix Household** on the menu, you will need to use a different TV device. 3. Select **Update Netflix Household**. 4. Select **Send Email** or **Send Text**. A verification link will be sent to the account email address or phone number. Verification links expire after 15 minutes. > **NOTE:** If you have not added a phone number or email address to your account, you will only see one option. 5. If you do not receive the verification link by email or text, click **Resend Email** or **Resend Text**. Or you can choose another option. 6. Select **Yes, This Was Me** in the email, or tap the link in the text message, then **Confirm Update** to continue. 7. You will receive a confirmation email and your TV will refresh so you can start watching. We use information such as IP addresses, device IDs, and account activity to determine whether a device signed into your account is part of your Netflix Household. Source: https://help.netflix.com/en/node/128339/us

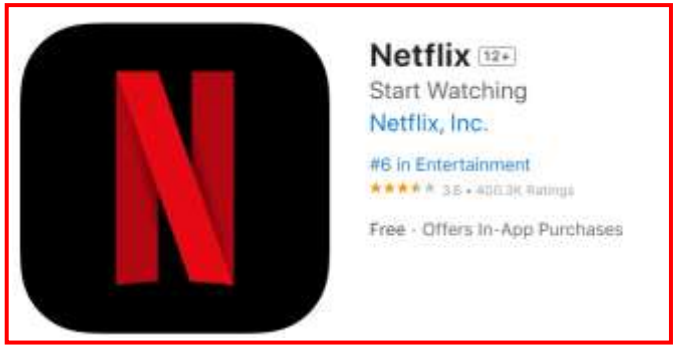
	A Netflix account is meant to be shared by people who live together in one household. People who are not in your household will need to sign up for their own account to watch Netflix. You can manage who uses your account by setting a Netflix Household. A Netflix Household is a collection of the devices connected to the internet at the main place you watch Netflix. A Netflix Household can be set using a TV device. All other devices that use your Netflix account on the same internet connection as this TV will automatically be part of your Netflix Household. Source: https://help.netflix.com/en/node/124925/us • **Device and network information:** We collect information about your computer or other Netflix capable devices you might use to access our service (such as smart TVs, mobile devices, set top boxes, gaming systems, and other streaming media devices), your network, and network devices. The information includes: ◦ device IDs or other unique identifiers, including for your network devices (such as your router), and devices that are Netflix capable on your network; ◦ IP addresses (which can be used to tell us the general location of your device, such as your city, state/province, and postal code); **From other sources:** We may collect the following categories of personal information about you from other ources: personal details, payment details, and device and network information. These sources include: ◦ **Service Providers** such as vendors, agents, and contractors that collect or provide personal information to Netflix in connection with services they perform on our behalf ("**Service Providers**"). This may include Service Providers such as those that: Source: https://help.netflix.com/legal/privacy#categories

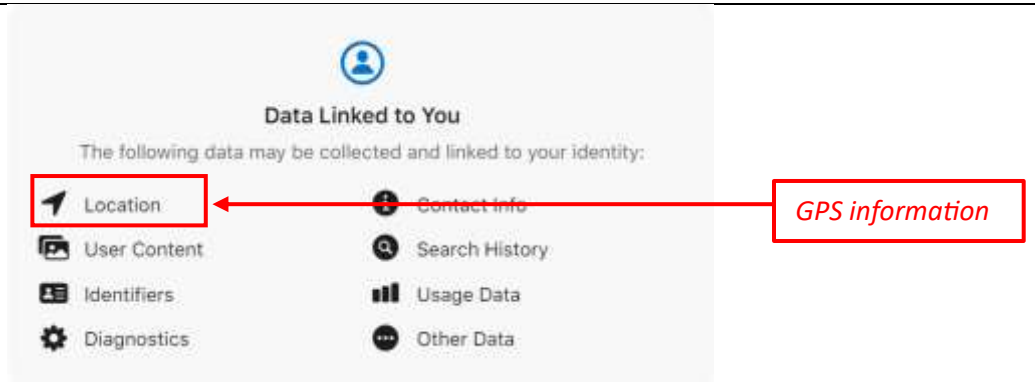
receiving at the processor, from the client computing device via the network interface, a request for access to the data, the request containing a client location of the client computing device, the client location including global positioning system (GPS) coordinates;	***Analyst Comment:*** *Receiving at the processor, from the client computing device (e.g., TV, Mobile, etc.) via the network interface, a request for access to the data, the request containing a client location of the client computing device, (e.g., TV, mobile, etc.) the client location including global positioning system (GPS) coordinates.* *When the user accesses Netflix from an Android or Apple device (e.g., TV or Mobile), the Netflix app collects the device location including GPS information. In addition to this, Netflix collects IP addresses and device IDs of the user's device as a part of location information.* **Client device**  Source: https://openconnect.netflix.com/Open-Connect-Overview.pdf

	Netflix entertains the world, providing a wide variety of TV series, films and games to hundreds of millions of members across the globe in over 30 languages. Netflix builds diversity, inclusion, equity, and a global outlook into everything it does, and by fostering a culture of courage, empathy, and curiosity, Netflix can move faster to develop new stories and better ways of sharing them with its members around the world. Netflix relies on AWS to help it innovate with speed and consistently deliver best-in-class entertainment. AWS provides Netflix with compute, storage, and infrastructure that allow the company to scale quickly, operate securely, and meet capacity needs anywhere in the world. Moreover, Netflix, a leading content producer, has used AWS to build a studio in the cloud. This virtual studio enables Netflix to engage top artistic talent, no matter the location, and Netflix artists and partners have the freedom to collaborate without technological or geographical barriers. Source: https://aws.amazon.com/solutions/case-studies/innovators/netflix/ 



Source: https://play.google.com/store/apps/details?id=com.netflix.mediaclient&hl=en



	 Source: https://apps.apple.com/us/app/netflix/id363590051 We use information such as IP addresses, device IDs, and account activity to determine whether a device signed into your account is part of your Netflix Household. Source: https://help.netflix.com/en/node/128339/us
comparing, at the processor, the client location to the approved geographic area;	***Analyst Comment:*** *Comparing, at the processor, the client location (e.g., IP address location) to the approved geographic area (e.g., location of the main place set by the user)* *Netflix compares the location (i.e., IP Address) of the user's device trying to access the service with the stored Netflix Household's main place's location information.* A Netflix account is meant to be shared by people who live together in one household. People who are not in your household will need to sign up for their own account to watch Netflix. You can manage who uses your account by setting a Netflix Household. A Netflix Household is a collection of the devices connected to the internet at the main place you watch Netflix. A Netflix Household can be set using a TV device. All other devices that use your Netflix account on the same internet connection as this TV will automatically be part of your Netflix Household.

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when the client location is within the approved geographic area, permitting access to the data for the client computing device via the network interface; and	***Analyst Comment**: When the client location (e.g., IP address location) is within the approved geographic area, permitting access to the data for the client computing device (e.g., TV, Mobile etc.,) via the network interface (e.g., Wi-Fi)* *Netflix grants access to its services if the user's device is connected to the household-approved IP address or is accessing the service from the designated primary location.* A Netflix account is meant to be shared by people who live together in one household. People who are not in your household will need to sign up for their own account to watch Netflix. You can manage who uses your account by setting a Netflix Household. A Netflix Household is a collection of the devices connected to the internet at the main place you watch Netflix. A Netflix Household can be set using a TV device. All other devices that use your Netflix account on the same internet connection as this TV will automatically be part of your Netflix Household. Source: https://help.netflix.com/en/node/124925/us **How to set or update a Netflix Household** A Netflix Household is a collection of devices connected to the internet at the main place you watch Netflix. A Netflix Household can be set using most TV devices. Devices that use your Netflix account on the same internet connection as this TV will automatically be part of your Netflix Household. We use information such as IP addresses, device IDs, and account activity to determine whether a device signed into your account is part of your Netflix Household. Source: https://help.netflix.com/en/node/128339/us

when the client location is not within the approved geographic area, denying the request.	***Analyst comment:*** *The client location is not within the approved geographic area, denying the request.* *If the user attempts to access the service from an unapproved location, Netflix denies the request, displays an error message stating, "This device is not part of the Netflix Household," and prompts the user to re-verify.* A Netflix account is meant to be shared by people who live together in one household. People who are not in your household will need to sign up for their own account to watch Netflix. You can manage who uses your account by setting a Netflix Household. A Netflix Household is a collection of the devices connected to the internet at the main place you watch Netflix. A Netflix Household can be set using a TV device. All other devices that use your Netflix account on the same internet connection as this TV will automatically be part of your Netflix Household. Source: https://help.netflix.com/en/node/124925/us We use information such as IP addresses, device IDs, and account activity to determine whether a device signed into your account is part of your Netflix Household. Source: https://help.netflix.com/en/node/128339/us **Netflix says 'Your device isn't part of the Netflix Household for this account'** Your device isn't part of the Netflix Household for this account **If you're using a mobile device or computer while on the go or away from home**, you will need to verify the device to watch temporarily. Source: https://help.netflix.com/en/node/133821/us

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